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Probing Fast Radio Burst Environments: Cross-Correlating Deep and Shallow FRB Surveys with KiDS DR5

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Abstract

Fast radio bursts (FRBs) are millisecond radio transients of extragalactic origin whose progenitor mechanisms remain unknown. Since FRBs at cosmological distances trace the large-scale structure, their angular clustering encodes the host-population bias, which depends on the progenitor channel. This thesis investigates how well this bias can be constrained by cross-correlating FRBs with the galaxy sample of KiDS-Legacy, the final data release of the Kilo-Degree Survey. Two progenitor scenarios, young magnetars and neutron star mergers formed through delayed channels, are each described by a redshift-dependent linear bias and evaluated for a shallow, CHIME-like FRB survey and a deep FRB survey representative of the Square Kilometre Array. The FRB auto-correlation alone is shot-noise dominated and does not constrain the bias. Adding the galaxy sample reduces the marginalized uncertainties by factors of 15 to 46 and improves the figure of merit by factors of 51 to 377, despite a sky area roughly 28 times smaller. With the current KiDS footprint, however, the two scenarios cannot yet be distinguished. A forecast for the Legacy Survey of Space and Time (LSST) shows that upcoming galaxy surveys will substantially improve the prospects of discriminating between a prompt and a delayed origin of FRBs.

Kurzfassung

Schnelle Radioblitze (engl. Fast Radio Bursts, FRBs) sind extragalaktische Radiotransienten mit Millisekundendauer, deren Ursprung bislang unbekannt ist. Da FRBs mit kosmologischen Entfernungen der großskaligen Struktur folgen, enthält ihre Winkelkorrelation Informationen über den Bias der Wirtspopulation, der vom Entstehungskanal abhängt. In dieser Arbeit wird eine Vorhersage-Pipeline entwickelt, um abzuschätzen, wie gut sich dieser Bias durch Kreuzkorrelation von FRBs mit der Galaxienstichprobe von KiDS-Legacy, dem finalen Datensatz des Kilo-Degree Survey, einschränken lässt. Zwei Entstehungsszenarien, junge Magnetare und Verschmelzungen von Neutronensternen aus verzögerten Kanälen, werden jeweils durch einen rotverschiebungsabhängigen linearen Bias beschrieben und für eine flache, CHIME-ähnliche sowie eine tiefe FRB-Durchmusterung nach dem Vorbild des Square Kilometre Array untersucht. Die FRB-Autokorrelation allein ist vom Schrotrauschen dominiert und schränkt den Bias nicht ein. Durch die Hinzunahme der Galaxien sinken die marginalisierten Unsicherheiten um Faktoren von 15 bis 46, und die Figure of Merit steigt um Faktoren von 51 bis 377, obwohl die Himmelsfläche etwa 28-mal kleiner ist. Mit der aktuellen KiDS-Fläche lassen sich die beiden Szenarien jedoch noch nicht unterscheiden. Eine Vorhersage für den Legacy Survey of Space and Time (LSST) zeigt, dass kommende Galaxiendurchmusterungen die Aussichten auf eine Unterscheidung zwischen prompten und verzögerten FRB-Ursprüngen erheblich verbessern werden.

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1 Introduction

Fast radio bursts (FRBs) are bright radio transients with millisecond durations that originate at cosmological distances [43]. Since the first burst was found in archival pulsar survey data [31], detections have grown rapidly, driven by the Canadian Hydrogen Intensity Mapping Experiment (CHIME) [8], whose second catalog contains over 3600 unique sources [9]. Beyond the open questions regarding their nature, FRBs are emerging as a novel cosmological probe. Their dispersion measure (DM) traces the diffuse ionized baryons, which are otherwise difficult to observe [35], and has been used to measure the cosmic baryon density [33] and the Hubble constant [14]. Larger samples could also constrain primordial non-Gaussianity [50] and baryonic feedback [54].

The astrophysical origin of FRBs, however, remains unresolved. A burst from the Galactic magnetar SGR 1935+2154 established a link to magnetars [7], whereas FRBs in old stellar environments point toward additional, delayed formation channels [59, 44]. Identifying individual host galaxies requires arcsecond localizations, which exist for only a small fraction of the population [49]. A statistical alternative exploits that FRBs trace the large-scale structure [34], so that their angular clustering encodes the bias of the host population and thereby the progenitor channel [55]. As FRB samples are sparse, their auto-correlation is dominated by shot noise, which cross-correlation with dense galaxy samples can mitigate. Such an FRB×galaxy cross-correlation has already been detected in the first CHIME/FRB catalog [48]. Photometric galaxy surveys like the Kilo-Degree Survey (KiDS), whose final data release KiDS-Legacy provides calibrated redshift distributions in six tomographic bins, are particularly well suited for this purpose [61, 62].

In this thesis, a forecasting pipeline is developed to assess how well the angular cross-correlation between FRBs and KiDS-Legacy galaxies can constrain the clustering bias of the FRB host population. Two progenitor scenarios are considered, namely young magnetars promptly following the cosmic star-formation history and neutron stars formed through delayed channels, each with a redshift-dependent linear bias. Both scenarios are evaluated for a shallow, CHIME-like FRB survey and a deep FRB survey modeled on the Square Kilometre Array (SKA) [16]. Angular auto- and cross-power spectra are computed in the Limber approximation and propagated into a Fisher forecast, which quantifies the gain of the joint analysis over the FRB auto-correlation alone and the prospects of distinguishing the two progenitor scenarios.

The structure of this thesis is as follows: Section 2 reviews the cosmological background, the properties and possible progenitors of FRBs, and the statistical description of the large-scale structure. Section 3 describes the modeling of both tracer populations and the computation of the angular power spectra. The Fisher forecast is presented and discussed in Section 4, while Section 5 summarizes the findings and gives an outlook on future, wider galaxy surveys.

2 Theoretical Background

2.1 Cosmological Background

Modern cosmology is founded on the cosmological principle, which states that the Universe is homogeneous and isotropic on large scales. Homogeneity implies that the large-scale properties of the Universe are statistically uniform across different locations, whereas isotropy means that no direction is preferred. Although introduced as an assumption, the principle is testable, and current evidence strongly supports it on the largest observable scales [53, 28].

Based on the cosmological principle, the Universe can be described by the Friedmann–Lemaître–Robertson–Walker (FLRW) metric

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2)\right], \tag{2.1}$$

with the spatial comoving coordinates (r, θ, ϕ), the speed of light c, and the cosmic time t [6]. The parameter k describes the curvature of the Universe, which can take values of -1, 0, or $+1$ for open, flat, and closed geometries, respectively. The scale factor $a(t)$ describes the relative expansion of the Universe and is conventionally normalized to its present value, $a(t_0) = 1$.

At earlier cosmic times, the scale factor satisfies $a(t) < 1$, whereas in an expanding universe it increases with time and can exceed unity in the future. A static universe would correspond to $\dot{a}(t) = 0$, resulting in a vanishing Hubble parameter

$$H(t) = \frac{\dot{a}(t)}{a(t)}, \tag{2.2}$$

which describes the relative change of the scale factor with cosmic time [20]. At the present time $H_0 = H(t_0)$ is referred to as the Hubble constant [6].

The expansion of space introduces a distinction between physical distances and comoving distances. While the physical distance between two objects changes with time due to the expansion of the Universe, the comoving distance remains constant for objects that follow the general expansion of space. The physical or so-called proper distance is related to the comoving distance χ by

$$D(t) = a(t)\chi\,. \tag{2.3}$$

Thus, comoving coordinates account for the effect of the cosmic expansion and provide a convenient description of the large-scale structure of the Universe [21].

The expansion of the Universe can be observed through the cosmological redshift. Due to the expansion of space, the wavelength of photons emitted by distant objects is stretched during their propagation. The redshift is defined as

$$z = \frac{\lambda - \lambda_\mathrm{e}}{\lambda_\mathrm{e}}\,, \tag{2.4}$$

where λ_e is the emitted wavelength and λ is the observed wavelength. The redshift is directly related to the scale factor through

$$1 + z = \frac{a(t_0)}{a(t_\mathrm{e})}\,, \tag{2.5}$$

with t_e denoting the time of emission. Consequently, the observed redshift provides information about the expansion history of the Universe and can be used to determine cosmological distances. The line-of-sight comoving distance to an object at redshift z is given by

$$\chi(z) = \chi_\mathrm{H} \int_0^z \frac{\mathrm{d}z'}{E(a(z'))}\,, \tag{2.6}$$

where $\chi_\mathrm{H} = c/H_0$ is the Hubble distance and $E(a) = H(a)/H_0$ is the dimensionless Hubble parameter [18].

2.2 Friedmann Equations and Cosmic Expansion

Following the cosmological principle, the matter and energy content of the Universe can be described by a perfect-fluid energy-momentum tensor. In combination with the FLRW metric, Einstein's field equations reduce to the Friedmann equations, which determine the evolution of the scale factor $a(t)$,

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda}{3}\,, \tag{2.7}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda}{3}\,. \tag{2.8}$$

Here ρ and p denote the energy density and pressure of the cosmic fluid, Λ is the cosmological constant, and G is the gravitational constant [20].

The critical density at the present time is defined as the total energy density for which the Universe is spatially flat ($k = 0$) and is given by

$$\rho_{\mathrm{c}} = \frac{3H_0^2}{8\pi G}\,. \tag{2.9}$$

The ratio between the density of a cosmological component and the critical density defines the corresponding density parameter,

$$\Omega_i = \frac{\rho_i}{\rho_{\mathrm{c}}}\,, \tag{2.10}$$

with Ω_{m}, Ω_{r}, Ω_Λ, and Ω_k denoting the density parameters of matter, radiation, vacuum (dark) energy, and curvature, respectively.

Consequently, the Friedmann equation can be expressed in terms of the density parameters as [2]

$$H^2 = H_0^2 \left[\Omega_{\mathrm{r}} a^{-4} + \Omega_{\mathrm{m}} a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda\right]\,. \tag{2.11}$$

2.3 The ΛCDM Model

The current standard model of cosmology is the so-called ΛCDM model. It describes the evolution of the Universe assuming the presence of cold dark matter (CDM) and a cosmological constant Λ, which is associated with dark energy and the accelerated expansion of the Universe. The model is in good agreement with a wide range of cosmological observations [41]. For the late-time Universe (low redshift) probed by extragalactic sources, the radiation density is negligible and the expansion history can be approximated by the contributions from matter, curvature, and dark energy. Assuming a spatially flat universe, the curvature term also vanishes, yielding [13]

$$E(z) \approx \sqrt{\Omega_{\mathrm{m},0}(1+z)^3 + \Omega_{\Lambda,0}}\,. \tag{2.12}$$

2.4 Fast Radio Bursts

FRBs are bright, transient radio pulses with durations of milliseconds or less and peak flux densities ranging from 50 mJy to 100 Jy [43]. They have been detected across a frequency range of 110 MHz [46] to 8 GHz [66].

Furthermore, the implied isotropic energies of FRBs are within a few orders of magnitude of prompt gamma-ray bursts and supernova explosions, suggesting that FRBs may originate from highly energetic astrophysical processes [43, 64]. Consequently, a broad range of progenitor models has been proposed, including neutron stars, magnetars, white dwarfs, black holes, ultra-luminous X-ray sources, active galactic nuclei, and more exotic scenarios such as superconducting cosmic strings [44].

Although a single millisecond burst releases as much energy as the Sun radiates in three days [43], the sources lie at cosmological distances, and the radio emission is diluted over an immense area as it propagates through the expanding Universe. The signal received on Earth is consequently roughly a thousand times weaker than that of a mobile phone on the Moon [5], as estimated in Section A.1.

The signal from FRBs is dispersed as it propagates through the ionized interstellar and intergalactic medium, causing lower-frequency components to arrive later than higher-frequency components. This dispersion is quantified by the DM, which follows from the measured arrival-time delay as

$$\mathrm{DM} = \frac{2\pi m_\mathrm{e} c}{e^2} \Delta t \left(\frac{1}{\nu_\mathrm{low}^2} - \frac{1}{\nu_\mathrm{high}^2} \right)^{-1} , \tag{2.13}$$

where ν_low and ν_high are the lower and higher observing frequencies, Δt is the time delay between the arrival of the two frequency components, c is the speed of light, e is the elementary charge, and m_e is the electron mass. The DM measures the integrated column density of free electrons between the source and the observer,

$$\mathrm{DM} \propto \int_0^L \mathrm{d}s \, n_\mathrm{e}(s) \, , \tag{2.14}$$

where n_e represents the free electron density, s the path length along the line of sight, and L the distance between the source and the observer [43].

The extragalactic origin of FRBs is inferred from their large DM values, which exceed the expected contribution from the Galactic interstellar medium along the corresponding line of sight. For a source located within the Milky Way, the values predicted by Galactic electron-density models typically do not exceed about $50\,\mathrm{pc\,cm^{-3}}$, whereas observed FRBs often exhibit considerably larger values of $1000\,\mathrm{pc\,cm^{-3}}$ and more [50].

The first FRB, later named the Lorimer Burst, was identified in archival pulsar survey data acquired in 2001 and published in 2007 [31]. Initially, only a small number of FRBs had been detected, leading to the assumption that they represented rare, singular events. The situation changed decisively with the commissioning of CHIME in 2018, whose wide field of view drove an exponential growth in detections and increased the known FRB population by more than an order of magnitude [8]. The latest CHIME/FRB

catalog contains 4539 detected burst events originating from 3641 unique FRB sources, including 981 bursts from 83 known repeating sources [9]. As shown in Figure 2.1, the known FRB sources exhibit an approximately isotropic distribution across the sky, consistent with an extragalactic population and suggesting no preferred spatial direction.

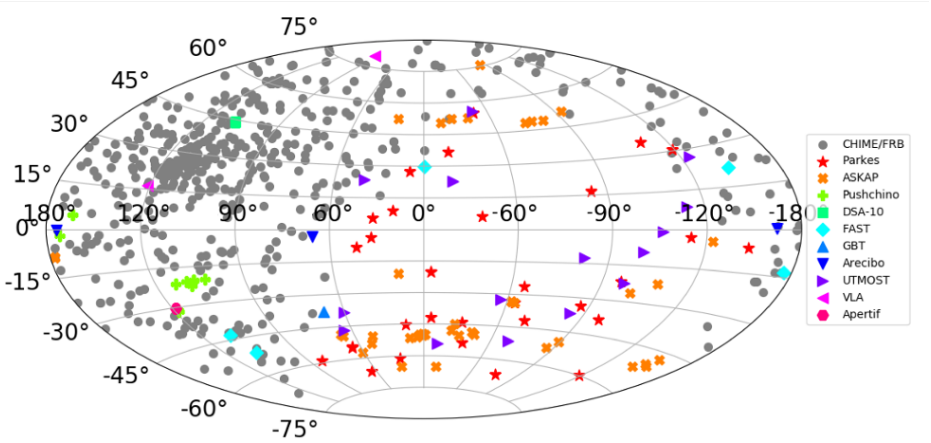


Figure 2.1: Aitoff projection of the known FRB population in Galactic coordinates based on observations available up to 2022. The sample is predominantly composed of CHIME/FRB detections, whose non-uniform sensitivity and sky coverage influence the observed distribution. Adapted from [44].

2.5 Progenitor Mechanisms

While the majority of FRBs have been observed as one-off events, a subset of sources has been found to emit multiple bursts over time. The existence of these so-called repeaters places a fundamental constraint on emission physics: the central engine must be capable of sustaining burst activity over timescales of years, which directly rules out any cataclysmic, single-event progenitor mechanism for these sources. Proposed progenitor models can therefore be broadly divided into catastrophic, one-off events, which can account only for non-repeating FRBs, and non-catastrophic scenarios involving long-lived central engines, which allow for multiple burst episodes over extended periods [43].

Some repeating FRBs also exhibit periodic modulation of their activity windows, as observed in FRB 20180916B [39], suggesting either a binary system geometry or precession of the emitting object. Among the non-catastrophic scenarios, compact stellar remnants, in particular neutron stars and magnetars, are considered the most promising candidates, supported by both theoretical models and an increasing body of observational evidence [43]. Whether all FRBs are intrinsically capable of repeating remains an open question [44].

Magnetars are neutron stars distinguished by surface magnetic fields many orders of magnitude stronger than those of ordinary radio pulsars, making them the most strongly magnetized stellar remnants known [43]. They arise primarily through the core collapse of massive stars in supernova explosions and are therefore intrinsically young objects that are preferentially found in or near regions of active star formation and whose formation rate is expected to closely mirror the cosmic star-formation history (SFH) [42]. The landmark detection in April 2020 of an FRB-like burst from the Galactic magnetar SGR 1935+2154 provided the most direct empirical link between magnetars and FRBs [7]. Population synthesis work using the CHIME/FRB catalog finds that the hypothesis of FRBs tracking the SFH is not ruled out by current data, with a small delay preferred, consistent with a scenario in which young magnetars formed through core-collapse supernovae serve as the dominant progenitor class [36].

Neutron star mergers formed through delayed channels represent a second important progenitor class, motivated by FRBs localized to environments inconsistent with active star formation. FRB 20180924A and FRB 20190523A, found in massive, low-star-formation-rate galaxies with large offsets from the host center, display environments similar to those of short gamma-ray burst hosts associated with mergers of binary neutron stars or of neutron star–black hole binaries [59]. The globular cluster localization of FRB 20200120E reinforces this point, as such an old environment is generally not associated with recent star formation. Isolated core-collapse supernovae occur within 3–48 Myr after a burst of star formation, whereas binary interactions can delay some core-collapse events to about 200 Myr [63]. Accretion-induced collapse and compact-object merger channels exhibit much broader delay-time distributions extending to several gigayears [52]. As a consequence, this progenitor population does not closely track the cosmic SFH but evolves more slowly with redshift [30].

Although the two classes also predict distinct redshift distributions [30], this work encodes their difference solely in a redshift-dependent bias $b(z)$, which reflects the host environments described above. The parameterization and specific values are detailed in Section 3.4.

2.6 Structure Formation

Within the ΛCDM framework, the large-scale structure of the Universe emerged from the gravitational amplification of small primordial density perturbations. According to the gravitational instability paradigm, quantum fluctuations seeded during an inflationary phase were stretched to cosmological scales and subsequently grew into the filaments, voids, and galaxy clusters observed today [3, 51].

These perturbations are described by the density contrast, defined as the fractional deviation of the local matter density $\rho(\mathbf{x})$ at comoving position $\mathbf{x}$ from the mean cosmic matter density $\bar{\rho}$,

$$\delta(\mathbf{x}) := \frac{\rho(\mathbf{x}) - \bar{\rho}}{\bar{\rho}} \,. \tag{2.15}$$

As long as $|\delta| \ll 1$, the evolution is linear and all Fourier modes grow at the same rate,

$$\delta(\mathbf{x}, z) = D_{+}(z)\, \delta(\mathbf{x}, 0) \,, \tag{2.16}$$

where the linear growth factor is normalized to $D_{+}(0) = 1$. The overall amplitude of matter fluctuations at the present time is conventionally parameterized by σ_8 [2].

Galaxies and other astrophysical tracers, such as the hosts of FRBs, do not trace the matter distribution one-to-one, since they form preferentially in the high-density peaks of the matter field. On large scales, the density contrast δ_{t} of a tracer population is related to that of the matter by the linear bias b,

$$\delta_{\mathrm{t}}(\mathbf{x}, z) = b(z)\, \delta(\mathbf{x}, z) \,, \tag{2.17}$$

where $b > 1$ indicates a tracer that is more strongly clustered than the matter [11].

2.7 Correlation Functions and Angular Power Spectrum

The statistical properties of the large-scale matter distribution are quantified through correlation functions and their Fourier-space counterparts, the power spectra. The 2-point correlation function of the density contrast is defined as

$$\xi(\mathbf{y}) := \langle \delta(\mathbf{x})\, \delta(\mathbf{x} + \mathbf{y}) \rangle \,, \tag{2.18}$$

where the angle brackets denote an ensemble average. Statistical homogeneity implies that ξ depends only on the separation $\mathbf{y}$ and not on the absolute position $\mathbf{x}$, while statistical isotropy further implies that it depends only on the magnitude $y = |\mathbf{y}|$.

For a discrete tracer population, the 2-point correlation function can be interpreted as describing the excess probability of finding a pair of tracers at a given separation relative to an unclustered random distribution. The Fourier-transform pair of $\delta(\mathbf{x})$ and $\hat{\delta}(\mathbf{k})$ is defined according to the convention

$$\hat{\delta}(\mathbf{k}) = \int \mathrm{d}^3x\, \delta(\mathbf{x})\, \mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{x}} \,; \qquad \delta(\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \hat{\delta}(\mathbf{k})\, \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} \,. \tag{2.19}$$

The 3-dimensional matter power spectrum is then defined through

$$\left\langle \hat{\delta}(\mathbf{k})\, \hat{\delta}^{*}(\mathbf{k}') \right\rangle = (2\pi)^3 \delta_{\mathrm{D}}(\mathbf{k}-\mathbf{k}') P(k)\,, \tag{2.20}$$

where δ_{D} denotes the Dirac delta function, reflecting statistical homogeneity in Fourier space. Statistical isotropy implies that the power spectrum depends only on the magnitude $k = |\mathbf{k}|$ [2].

For projected tracer fields on the celestial sphere, spherical harmonics provide a natural description on large angular scales, where the flat-sky approximation is not generally adequate. A scalar field $f(\hat{\mathbf{n}})$, where $\hat{\mathbf{n}}$ denotes the line-of-sight direction, can be expanded as $f(\hat{\mathbf{n}}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{\mathbf{n}})$, with harmonic coefficients $a_{\ell m} = \int \mathrm{d}\Omega\, f(\hat{\mathbf{n}}) Y^{*}_{\ell m}(\hat{\mathbf{n}})$ [48].

Using the spherical-harmonic expansion together with the Fourier representation of the 3-dimensional fields, the full-sky angular cross-power spectrum can be written as

$$C_\ell^{AB} = \frac{2}{\pi} \int_0^\infty k^2\, \mathrm{d}k \int \mathrm{d}\chi\, W^A(\chi)\, j_\ell(k\chi) \int \mathrm{d}\chi'\, W^B(\chi')\, j_\ell(k\chi')\, P_{AB}(k;\chi,\chi')\,, \tag{2.21}$$

where j_ℓ denotes the spherical Bessel function and P_{AB} is the 3-dimensional cross-power spectrum of the two tracer fields. The Limber approximation states that for multipoles of order 10 and larger, the double integral collapses to a single line-of-sight integral [29],

$$C_\ell^{AB} \approx \int \frac{\mathrm{d}\chi}{\chi^2}\, W^A(\chi)\, W^B(\chi)\, P_{AB}\left(k = \frac{\ell+1/2}{\chi}, z(\chi)\right)\,. \tag{2.22}$$

The use of $\ell + 1/2$, rather than simply ℓ, provides a more accurate version of the Limber approximation, particularly at low multipoles [32]. The characteristic angular scale associated with a given multipole ℓ is approximately $\theta \sim 1/\ell$ [48].

For linearly biased tracers, Equation 2.17 implies

$$P_{AB}(k,z) = b_A(z)\, b_B(z)\, P(k,z)\,, \tag{2.23}$$

so that the bias factors can be absorbed into the radial kernels W^A and W^B. The angular cross-power spectrum therefore depends on the radial overlap of the two tracer populations, their relation to the underlying matter density field through their bias factors, and the shape and redshift evolution of the underlying matter power spectrum.

Consequently, the FRB×galaxy angular cross-power spectrum provides a sensitive probe of the spatial relation between FRBs and galaxies and can be used to constrain the redshift distribution, clustering properties, and host-population characteristics of FRBs [48, 55].

3 Methodology

This chapter describes the analysis pipeline developed in this thesis to model the angular auto- and cross-power spectra of FRBs and photometrically selected galaxies. The pipeline is implemented in `Python` [47] and follows the theoretical formalism established in Section 2. All numerical operations are carried out using `NumPy` [15] and `SciPy` [58], while all figures are generated with `Matplotlib` [19].

3.1 Fiducial Cosmology and Numerical Grids

Throughout this thesis, a spatially flat ΛCDM cosmology as introduced in Section 2.3 is assumed. The fiducial parameter values are taken from the Planck 2018 baseline analysis of the combined TT,TE,EE+lowE+lensing likelihood [45] and are summarized in Table 3.1.

Table 3.1: Fiducial cosmological parameters assumed throughout this thesis. The values correspond to the Planck 2018 baseline TT,TE,EE+lowE+lensing results [45].

Parameter	Symbol	Value
Hubble constant	H_0 [km s^{-1} Mpc^{-1}]	67.36
Total matter density	$\Omega_{\mathrm{m},0}$	0.3153
Baryon density	$\Omega_{\mathrm{b},0}$	0.0493
Fluctuation amplitude	σ_8	0.8111
Scalar spectral index	n_{s}	0.9649
CMB temperature	$T_{\mathrm{CMB},0}$ [K]	2.7255

The corresponding background model is realized as a `FlatLambdaCDM` instance of `astropy.cosmology` [1], from which the line-of-sight comoving distance $\chi(z)$ and the Hubble rate $H(z)$ are evaluated numerically following the relations given in Section 2.1.

All redshift-dependent quantities are evaluated on a common logarithmically spaced grid of 1000 points covering $z \in [0.01, 5]$. Angular power spectra are evaluated on a dense grid of integer multipoles $\ell \in [10, 1000]$. The lower limit is set by the validity range of the Limber approximation discussed in Section 2.7, which requires multipoles of order 10 or larger [29, 32], while the upper limit restricts the analysis to angular scales for which the adopted nonlinear matter power spectrum prescription remains sufficiently reliable.

3.2 Nonlinear Matter Power Spectrum

The 3-dimensional matter power spectrum $P(k, z)$ introduced in Section 2.7 is computed using the `hmf` package [38]. The resulting interpolator is shown in Figure 3.1.

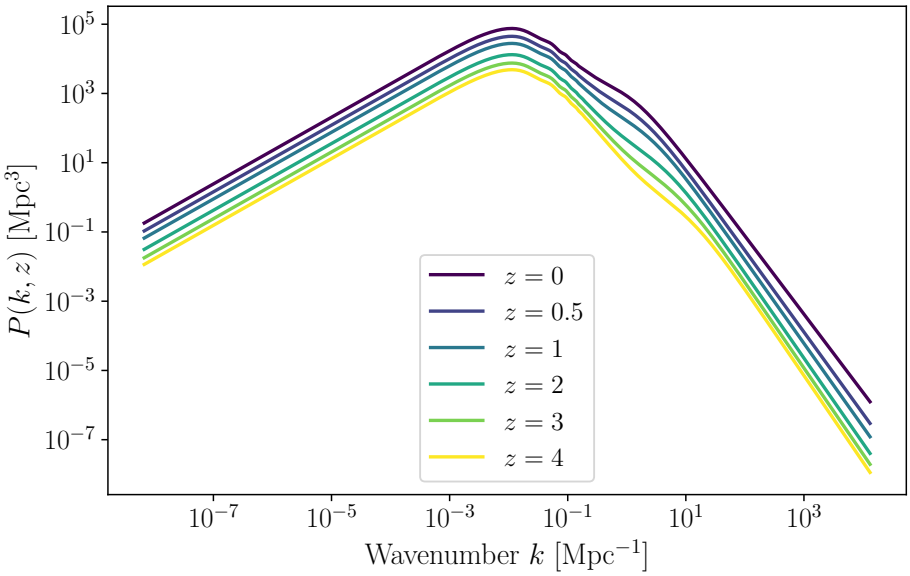


Figure 3.1: Nonlinear matter power spectrum $P(k, z)$ evaluated at six representative redshifts.

3.3 Redshift Distributions

3.3.1 FRB Redshift Distribution

Since the FRB population considered in this thesis is not associated with individually measured redshifts, its radial selection function has to be modeled. Following Reischke, Hagstotz, and Lilow [50], the normalized FRB redshift distribution is assumed to take the form

$$n(z) \propto z^2 \exp(-\alpha z)\,, \tag{3.1}$$

where the parameter α controls the steepness of the high-redshift tail and thereby the effective depth of the survey. The z^2 prefactor reflects the growth of the comoving volume element at low redshift, whereas the exponential cutoff encodes the decreasing detectability of bursts at large distances. The distribution is normalized numerically on the redshift grid defined in Section 3.1, such that $\int n(z)\,\mathrm{d}z = 1$.

Two survey configurations are considered, following Reischke et al. [50], and summarized in Table 3.2. Both distributions are shown in Figure 3.2.

Table 3.2: FRB survey configurations adopted in this thesis. The parameters α and N_{tot} determine the redshift distribution and shot-noise level according to Equation 3.1 and Equation 3.9, respectively. The sky fraction f_{sky} is identical for both configurations.

Configuration	α	N_{tot}	f_{sky}
Shallow	3.5	5000	0.9
Deep	2.0	50 000	0.9

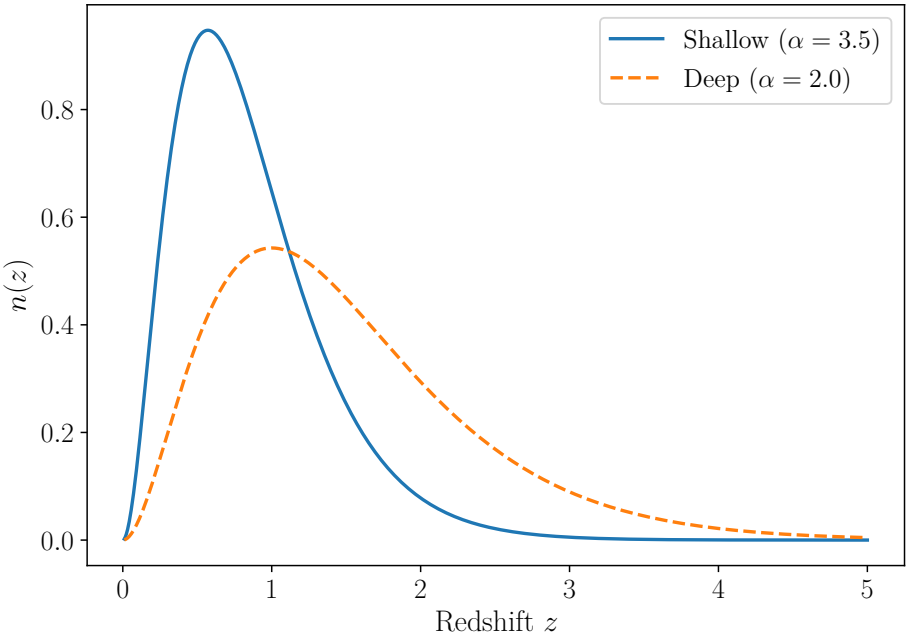


Figure 3.2: Normalized FRB redshift distributions given in Equation 3.1 for the shallow ($\alpha = 3.5$) and deep ($\alpha = 2.0$) survey configurations.

The corresponding comoving distance distribution has to be transformed accordingly. Conservation of the number of sources, $n(\chi)\,\mathrm{d}\chi = n(z)\,\mathrm{d}z$, yields

$$n(\chi) = n(z)\,\frac{\mathrm{d}z}{\mathrm{d}\chi}\,, \tag{3.2}$$

with the Jacobian $\mathrm{d}z/\mathrm{d}\chi = H(z)/c$ following from the definition of the comoving distance in Section 2.1.

3.3.2 Galaxy Redshift Distribution

KiDS is a wide-field optical imaging survey carried out with the 2.6 m VLT Survey Telescope at Paranal. It covers 1347 deg^2 and, combined with VIKING, provides nine-band optical-to-near-infrared photometry [24, 62]. The data used in this thesis are drawn from the fifth data release (DR5), KiDS-Legacy, which reaches a $5\,\sigma$ limiting magnitude of 24.8 in r with a median seeing of $0\farcs7$ [62, 27]. Photometric redshifts are calibrated with about 126 000 spectroscopic sources [61]. For the subsequent analysis, the redshift distributions of the six tomographic bins are normalized to unit integral. The raw tabulated distributions are shown in Figure 3.3.

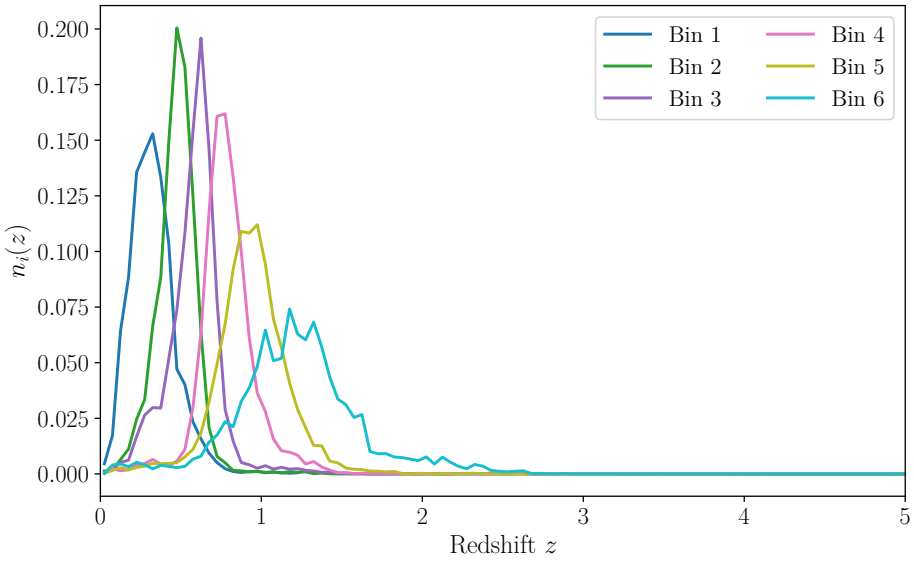


Figure 3.3: Raw, tabulated tomographic redshift distributions $n_i(z)$ of the six KiDS-Legacy galaxy bins as published by Wright et al. [61] and the KiDS Collaboration [26].

3.4 Tracer Bias Models

3.4.1 FRB Host Population Bias

As motivated in Section 2.5, the two progenitor classes are expected to trace the underlying matter distribution differently. This difference is encoded in a redshift-dependent linear bias of the power-law form

$$b(z) = b_0\,(1+z)^{\delta}\,, \tag{3.3}$$

where b_0 denotes the bias amplitude at $z = 0$ and the exponent δ parameterizes its redshift evolution [40]. The fiducial values adopted for the two progenitor classes are listed in Table 3.3 and shown in Figure 3.4.

Table 3.3: Fiducial parameters of the FRB host population bias model.

Progenitor class	b_0	δ
Magnetars	1.0	0.8
Neutron stars	1.5	0.2

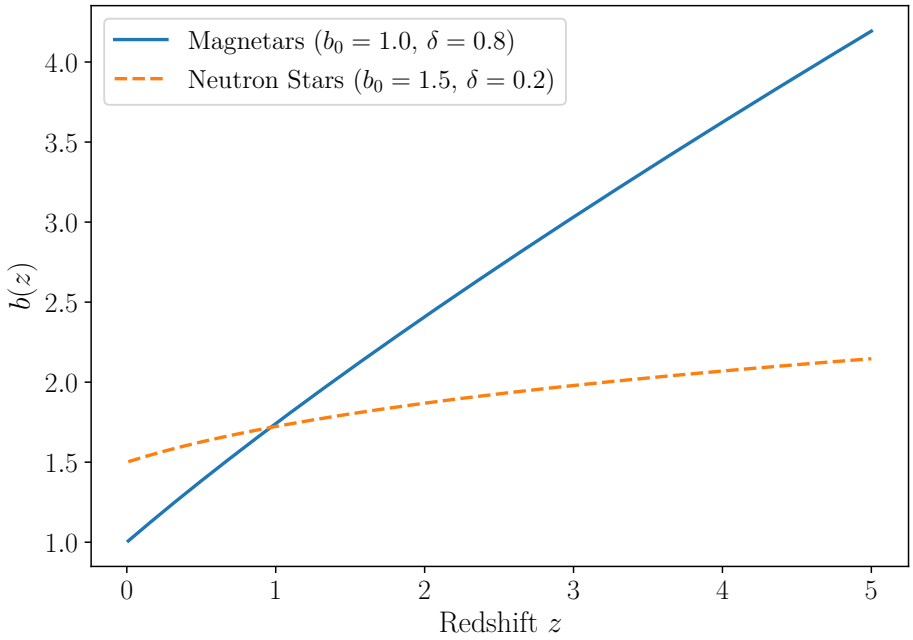


Figure 3.4: FRB host population linear bias models of Equation 3.3 for the two progenitor classes listed in Table 3.3.

3.4.2 Galaxy Bias

For the galaxy sample a constant-clustering bias model is adopted, in which the comoving clustering amplitude of the sample is assumed to remain constant over the redshift range spanned by each tomographic bin. Under this assumption the bias evolves inversely with the linear growth factor,

$$b_{\mathrm{g}}^{i} = \frac{0.95}{D_{+}(\langle z \rangle_{i})}\,, \tag{3.4}$$

which is a common prescription used in photometric survey forecasts [23]. Here $D_+(z)$ denotes the linear growth factor of Equation 2.16, evaluated with `hmf` [38], and $\langle z \rangle_i$ is the mean redshift of the i-th tomographic bin. The resulting mean redshifts and bias values are listed in Table 3.4.

Table 3.4: Properties of the six KiDS-Legacy tomographic galaxy bins. The mean redshift $\langle z \rangle_i$ is computed from the published redshift distributions and the bias b_{g}^i follows from Equation 3.4.

Bin	$\langle z \rangle_i$	b_{g}^i
1	0.335	1.135
2	0.477	1.221
3	0.587	1.291
4	0.789	1.423
5	0.940	1.525
6	1.224	1.721

3.5 Radial Weight Functions

The radial kernels $W^A(\chi)$ and $W^B(\chi)$ introduced in Section 2.7 weight the 3-dimensional density contrast along the line of sight. For a discrete tracer population whose density contrast is related to the matter density contrast through the linear bias of Equation 2.17, the tracer contribution to the radial kernel is proportional to the product of the bias and the normalized radial selection function [25].

For the FRB sample the weight function is therefore constructed on the redshift grid as

$$W_{\mathrm{FRB}}(z) = b(z)\, n(z)\,, \tag{3.5}$$

with $n(z)$ from Equation 3.1 and $b(z)$ from Equation 3.3. For the galaxy sample, the radial kernel of the i-th tomographic bin is given by

$$W_{\mathrm{gal}}^i(z) = b_{\mathrm{gal}}^i\, n_i(z)\,, \tag{3.6}$$

where b_{gal}^i is the galaxy bias and $n_i(z)$ is the normalized redshift distribution of the corresponding bin [37]. Both kernels are subsequently transformed to comoving distance using the Jacobian introduced in Section 3.3.1,

$$W(\chi) = W(z)\frac{H(z)}{c}\,, \tag{3.7}$$

which gives the radial kernels entering the projection integral.

3.6 Angular Power Spectrum Computation

The angular power spectra are computed from the Limber-approximated projection integral of Equation 2.22. For each multipole the integrand

$$\frac{W^A(\chi)\,W^B(\chi)}{\chi^2}\,P\!\left(k=\frac{\ell+1/2}{\chi},\,z(\chi)\right) \tag{3.8}$$

is evaluated on the comoving-distance grid corresponding to the redshift grid of Section 3.1 and integrated with the composite trapezoidal rule.

3.6.1 Auto-Correlation Analysis

For the auto-correlation of a single tracer population, the two kernels in Equation 2.22 coincide, $W^A = W^B$, and the projection integral reduces to a weighted line-of-sight average of the matter power spectrum. Auto-spectra are computed for both FRB survey configurations and both progenitor classes, as well as for each of the six tomographic galaxy bins.

In contrast to the cross-spectra discussed below, auto-spectra receive an additive Poissonian shot-noise contribution, because both tracers are discrete point processes. For a sample of mean projected number density $\bar{n}$ per steradian, this contribution is scale-independent and given by

$$N_\ell = \frac{1}{\bar{n}}\,. \tag{3.9}$$

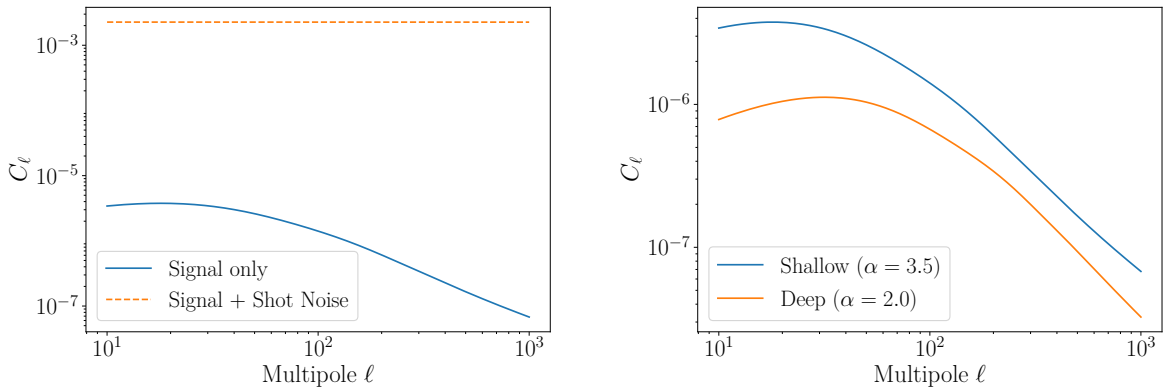
For the FRB samples the number density is given by

$$\bar{n}_{\mathrm{FRB}} = \frac{N_{\mathrm{tot}}}{4\pi f_{\mathrm{sky}}}\,, \tag{3.10}$$

following from the total number of detected bursts and the observed sky fraction in Table 3.2 [50]. For the galaxy bins, the mean number densities $\bar{n}$ per tomographic bin are taken from the KiDS data [26].

For the representative shallow FRB configuration, the shot noise exceeds the clustering signal by roughly three orders of magnitude across the entire multipole range, rendering the auto-correlation strongly noise-dominated for a sample of only 5×10^3 bursts (Figure 3.5a). This illustrates the limited constraining power of the FRB auto-spectrum

alone at current survey number densities and motivates its combination with a dense galaxy sample in a cross-correlation analysis.



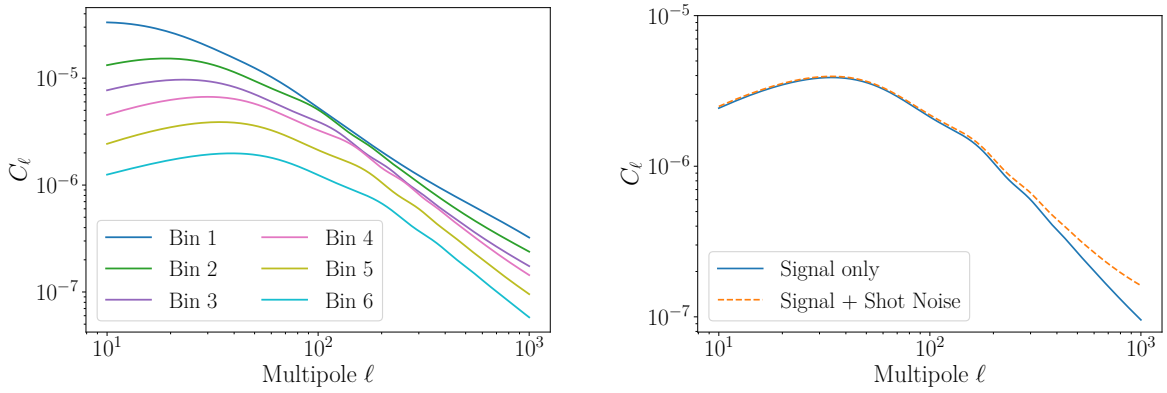
(a) Shallow survey, with and without shot noise.

(b) Shallow and deep surveys, signal only.

Figure 3.5: Auto-correlation angular power spectra of the magnetar FRB population.

The signal-only spectra show that the deep survey configuration yields a lower amplitude than the shallow one, as illustrated in Figure 3.5b. At higher redshifts, structure growth is less advanced, resulting in a matter power spectrum with a lower amplitude. Moreover, the geometric factor χ^{-2} in Equation 2.22 suppresses the contribution from sources at larger comoving distances. The same qualitative behavior is found for the delayed neutron-star population.

The galaxy auto-spectra of all six tomographic bins are shown in Figure 3.6a. The amplitude decreases systematically toward higher redshift, despite the increasing galaxy bias given by Equation 3.4.



(a) Signal-only spectra for the six tomographic bins.

(b) Signal and shot noise for bin 5.

Figure 3.6: Galaxy auto-correlation angular power spectra for the six KiDS-Legacy tomographic bins.

This is mainly due to the broader radial extent of the higher-redshift tomographic bins, as discussed above in the context of the FRB auto-correlation. The relative contribution of shot noise is illustrated in Figure 3.6b for bin 5, where the signal falls below the shot-noise level toward high multipoles.

3.6.2 Cross-Correlation Analysis

The angular cross-power spectrum between the FRB sample and the i-th tomographic galaxy bin is obtained from Equation 2.22 by inserting the two distinct kernels of Equation 3.5 and Equation 3.6. Cross-spectra are computed for all combinations of two progenitor classes, two survey configurations, and six tomographic bins, yielding $2 \times 2 \times 6 = 24$ spectra in total. Figure 3.7 contrasts the two survey configurations for the magnetar population in all six tomographic bins. Although the deep configuration has a larger FRB contribution at the highest redshifts, the sixth galaxy bin remains broad and receives substantial weight below the crossing redshift of the shallow and deep FRB distributions. Together with the low-redshift weighting of the Limber kernel, this allows the shallow cross-spectrum to remain larger even in the highest tomographic bin.

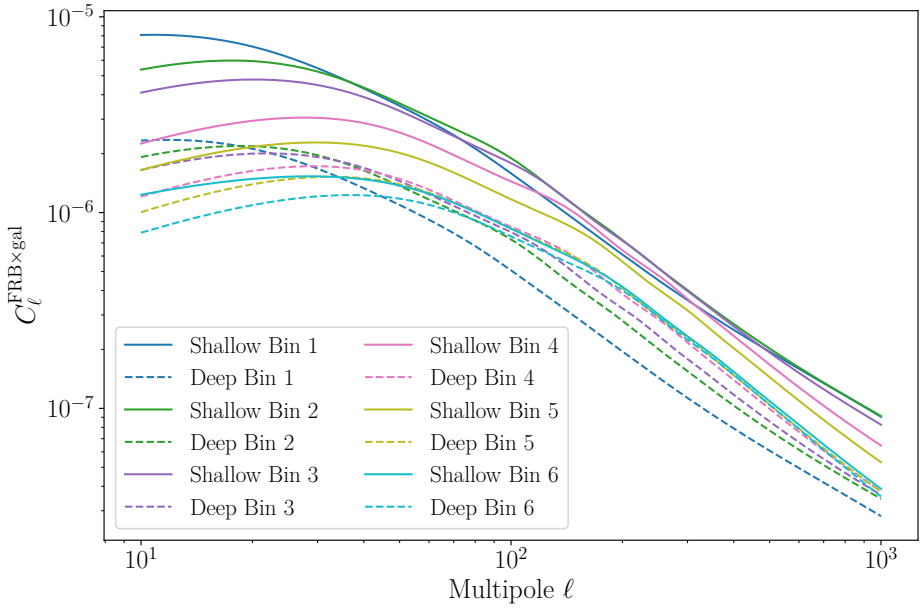


Figure 3.7: Comparison of the FRB×galaxy cross-power spectra of the shallow and deep survey configuration for the magnetar population, shown for all six tomographic galaxy bins.

4 Results

The angular power spectra derived in Section 3 represent theoretical predictions for a specific set of model parameters. To assess how well these parameters could be recovered from an actual measurement, the likelihood is assumed to be Gaussian around the fiducial model. Under this assumption, the expected parameter covariance follows directly from the Fisher information matrix.

4.1 Fisher Matrix

The Fisher matrix quantifies how sensitively the observables respond to variations in the model parameters and therefore how much information a given measurement carries about them. For a set of tracer fields whose harmonic coefficients are Gaussian distributed, the Fisher matrix of the angular power spectra C_ℓ is given by [56, 50]

$$F_{ij} = f_{\mathrm{sky}} \sum_{\ell=\ell_{\mathrm{min}}}^{\ell_{\mathrm{max}}} \frac{2\ell+1}{2} \,\mathrm{Tr}\left[C_\ell^{-1} \frac{\partial C_\ell}{\partial \theta_i} C_\ell^{-1} \frac{\partial C_\ell}{\partial \theta_j} \right], \tag{4.1}$$

where θ_i denotes the i-th model parameter, C_ℓ is the matrix of auto- and cross-power spectra over all tracer pairs, including the shot noise of Equation 3.9 in the auto-spectra. The trace runs over the tracer indices, and f_{sky} accounts for the fact that only a fraction of the celestial sphere is observed.

The derivatives entering Equation 4.1 are evaluated by central finite differencing,

$$\frac{\partial C_\ell}{\partial \theta_i} = \frac{C_\ell(\theta_i + h_i) - C_\ell(\theta_i - h_i)}{2h_i}, \tag{4.2}$$

with step sizes $h_i = 0.01\,\theta_i$. The two parameters considered here are the amplitude b_0 and the evolution exponent δ of the FRB host bias of Equation 3.3, with fiducial choices given in Table 3.3. All cosmological parameters are fixed to those listed in Table 3.1.

The FRB-only forecast adopts $f_{\mathrm{sky}} = 0.9$ as given in Table 3.2, whereas the joint forecast uses $f_{\mathrm{sky}} = 0.0327$, corresponding to the KiDS footprint of $1347\,\mathrm{deg}^2$ quoted in Section 3.3.2.

Inverting the Fisher matrix yields the parameter covariance $\mathrm{Cov} = F^{-1}$, whose diagonal entries define the marginalized uncertainties $\sigma_i = \sqrt{\mathrm{Cov}_{ii}}$ [10]. To compress the 2-dimensional constraint into a single number, the figure of merit (FoM)

$$\mathrm{FoM} = \frac{1}{\sqrt{\det \mathrm{Cov}}} \tag{4.3}$$

is used, which is inversely proportional to the area of the confidence ellipse, so that larger values correspond to tighter constraints [60].

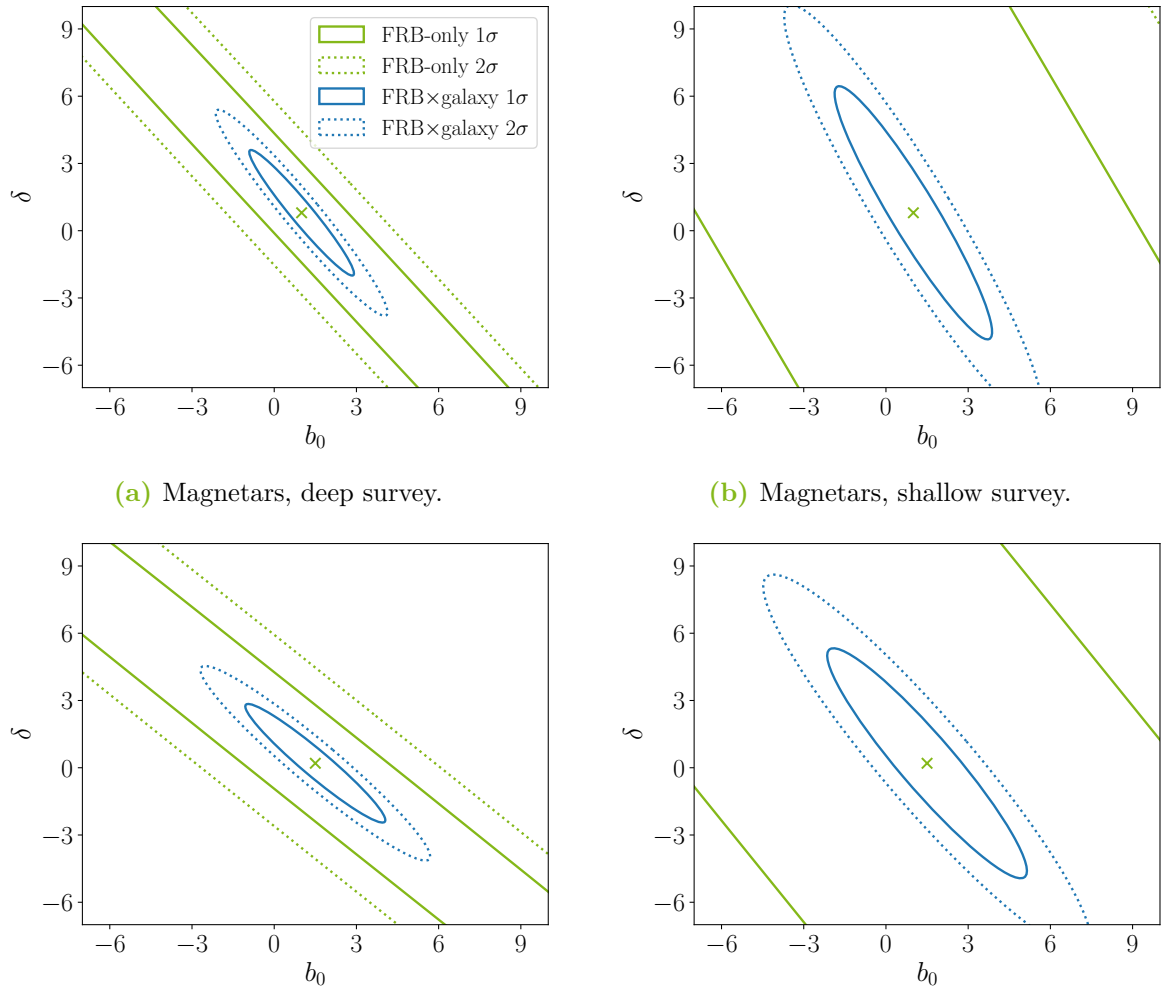
4.2 Forecast Constraints

The forecast is evaluated for all four combinations of progenitor models (see Table 3.3) and FRB survey configurations (see Table 3.2). The resulting confidence ellipses are shown in Figure 4.1, while the marginalized uncertainties and FoMs are listed in Table 4.1.

Table 4.1: Marginalized uncertainties and FoMs for the FRB bias parameters. The gain is the ratio of the joint to the FRB-only FoM and is computed from unrounded values.

		σ_{b_0}		σ_δ		FoM		
Progenitor	Configuration	FRB	FRB×gal	FRB	FRB×gal	FRB	FRB×gal	Gain
Magnetars	Deep	21	1.26	28	1.85	0.032	1.6	51
Magnetars	Shallow	83	1.90	172	3.73	0.001	0.4	377
Neutron stars	Deep	30	1.69	29	1.75	0.020	1.1	60
Neutron stars	Shallow	92	2.41	139	3.39	0.001	0.3	294

The FRB auto-correlation alone does not constrain the host bias. Marginalized uncertainties range from $\sigma_{b_0} = 21$ to 92 and $\sigma_\delta = 28$ to 172, and therefore exceed the fiducial values by one to two orders of magnitude. Adding the galaxy sample reduces these to $\sigma_{b_0} = 1.26$–2.41 and $\sigma_\delta = 1.85$–3.73. This corresponds to improvements by factors of 15–46 in the individual parameters and 51–377 in the FoM. The joint analysis is therefore not a marginal refinement but the reason a measurement becomes feasible in the first place.



(a) Magnetars, deep survey.

(b) Magnetars, shallow survey.

(c) Neutron stars, deep survey.

(d) Neutron stars, shallow survey.

Figure 4.1: Fisher forecast in the $b_0 - \delta$ plane for the two progenitor models and the two FRB survey configurations. Solid and dotted lines mark the 1σ and 2σ confidence regions obtained from the FRB auto-correlation alone (green) and from the joint analysis of the FRB sample with the six tomographic KiDS bins (blue), as given in the legend of panel (a). The green cross marks the fiducial parameter values. The FRB-only contours extend far beyond the plotted range in every panel.

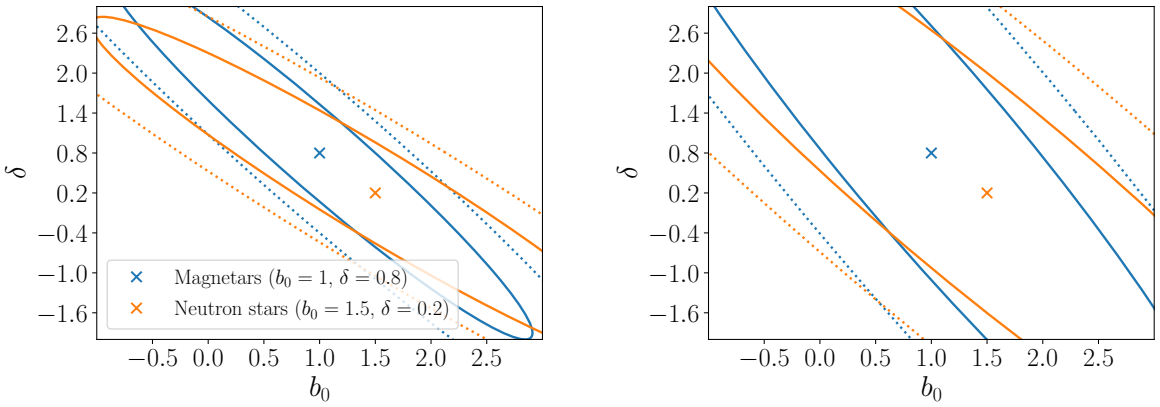
4.3 Discussion

Comparing the two survey configurations, the deep sample outperforms the shallow one in every case, as expected from its 10 times larger burst count and correspondingly lower shot noise. These two configurations are chosen to resemble realistic experiments, with the shallow survey being CHIME-like [9] and the deep survey SKA-like [16]. The comparison therefore doubles as an estimate of how much constraining power a next-

generation instrument would gain over the current generation. However, this advantage propagates differently into the two forecasts: in the FRB-only case the deep configuration improves σ_{b_0} by factors of 3 to 4, whereas in the joint case the gain is only about 1.4–1.5. The reason for that is the limited radial overlap between the FRB and galaxy sample, as around 40% of the deep-survey bursts lie at $z > 1.5$, beyond the bulk of the KiDS galaxies, and therefore add little to the cross-correlation.

A comparison of the two progenitor models requires some care, as their fiducial values differ. In absolute terms, the magnetar model yields tighter constraints. Relative to their respective fiducial amplitudes, however, both models are constrained similarly well, with $\sigma_{b_0}/b_0 = 1.3$ for the magnetar model and 1.1 for the neutron-star model in the deep configuration.

The more consequential question is whether the two models can be distinguished. To address this, Figure 4.2 overlays the joint contours of both progenitor models for each survey configuration. The fiducial separation of $\Delta b_0 = 0.5$ and $\Delta\delta = 0.6$ lies well within the 1σ contours of the joint forecast, so the contours overlap almost entirely. Hence, the cross-correlation with KiDS improves the measurement by roughly two orders of magnitude in the FoM but does not yet reach the precision required to discriminate between a prompt magnetar and a delayed neutron star origin.



(a) Deep survey configuration.

(b) Shallow survey configuration.

Figure 4.2: Joint FRB×galaxy forecast in the $b_0 - \delta$ plane for the deep (a) and shallow (b) FRB survey configurations, combined with the six tomographic KiDS bins. The contours of the magnetar model are shown in blue and those of the neutron-star model in orange. Solid lines mark the 1σ and dotted lines the 2σ confidence regions of the forecast. The crosses indicate the fiducial values of each model as given in the legend of panel (a).

5 Conclusion & Outlook

This thesis assessed whether the clustering of FRBs can constrain the bias of their host population, and to what extent cross-correlating them with a photometric galaxy survey can improve this measurement. A complete forecasting pipeline was developed that combines the nonlinear matter power spectrum with radial weight functions for modeled FRB samples and the six tomographic bins of the KiDS-Legacy data release to obtain angular auto- and cross-power spectra using the Limber approximation. Poissonian shot noise was added to the auto-spectra of both tracer populations, while the cross-spectrum was assumed to be free of shot noise. The resulting spectra were then propagated into a Fisher forecast for the FRB host-bias amplitude b_0 and evolution exponent δ.

The combination of the FRB sample with the KiDS galaxy survey yields strong constraints on the parameters of interest. The marginalized uncertainties are reduced by factors of 15 to 46, while the FoM improves by factors of 51 to 377, despite using a footprint roughly 28 times smaller than that assumed for the FRB-only analysis. These results demonstrate the strong constraining power gained from combining the FRB and galaxy samples. However, the current KiDS footprint is not yet sufficient to distinguish between a prompt magnetar channel and a delayed neutron-star channel, as the separation between the two fiducial models remains well within the 1σ contours.

5.1 Limitations

The analysis rests on some simplifying assumptions that may affect the forecast.

- The Limber approximation restricts the analysis to multipoles $\ell \geq 10$, excluding precisely the large angular scales.
- The Gaussian likelihood and purely Gaussian covariance neglect non-Gaussian contributions from nonlinear structure growth. The forecast uncertainties should therefore be interpreted as optimistic lower bounds rather than as realistic estimates.
- The bias is modeled as a power law, $b = b_0(1+z)^\delta$, which describes the locally observed FRB host population well at low redshift [4]. At higher redshift, the bias is expected to evolve more strongly and to depend on the host halo mass [40, 57]. The power law remains adequate here, as the constraints are dominated by $z \lesssim 1.5$.
- The FRB redshift distribution and fiducial bias values are model assumptions rather than measured quantities.

- The survey geometry is simplified by accounting for an incomplete sky coverage only through the f_{sky} prefactor, while a real analysis would require a full pseudo-spectrum treatment [17].

5.2 Outlook

The Fisher information of Equation 4.1 scales linearly with f_{sky}, so the most direct improvement is to replace the KiDS field with a wider galaxy survey. The Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory will image about $18\,000\,\mathrm{deg}^2$ of the southern sky [22], corresponding to $f_{\mathrm{sky}} = 0.4363$, roughly 13 times the KiDS area. Running the pipeline with the ten-bin Y10 lens sample of the LSST Dark Energy Science Collaboration shown in Figure A.1 [65] while keeping all other ingredients unchanged yields the contours shown in Figure A.2 and the numbers listed in Table A.1 and Table A.2.

The improvement is substantial: for the deep magnetar configuration, the uncertainty on b_0 drops from 1.26 to 0.42. The FoM increases by factors of 9–12, slightly less than the factor of 13 expected from the larger footprint alone, since the FoM scales linearly with f_{sky}. Per unit area, the LSST lens sample is thus slightly less informative than KiDS, most notably for the deep survey, because the LSST lens bins end at $z \approx 1.2$, whereas the sixth KiDS bin extends beyond $z = 1.5$ into the survey's high-redshift tail.

The separation between the two progenitor models, $\Delta b_0 = 0.5$ and $\Delta\delta = 0.6$, becomes comparable to the marginalized uncertainties in the deep configuration. This is illustrated in Figure A.3, the LSST counterpart of Figure 4.2.

In the deep configuration, the neutron-star fiducial now lies just outside the 1σ contour of the magnetar model, whereas the magnetar fiducial still falls inside the 1σ contour of the neutron-star model, and both remain within the 2σ regions.

In the shallow configuration, the two models still overlap at the 1σ level. A cross-correlation with an LSST-like sample would therefore substantially improve the prospects for distinguishing between a prompt and a delayed origin of the bursts, even though it would not yet allow a decisive discrimination.

Beyond a wider footprint, several extensions of the analysis are conceivable.

- Replacing the parametric redshift distribution of Equation 3.1 by the measured distribution of the second CHIME/FRB catalog [9], which already provides a sample of shallow-configuration size.
- Full likelihood sampling instead of the Fisher formalism, which would capture non-Gaussian posterior features and non-Gaussian covariance contributions.

- A scale-dependent halo-model description of the bias, linking it to host halo masses and extending the analysis to smaller scales

In conclusion, the results of this thesis demonstrate the strong potential of FRB×galaxy cross-correlation as a probe of FRB progenitors. Combining the FRB sample with a dense photometric galaxy sample efficiently recovers clustering information and enables meaningful constraints on the host bias with current survey footprints. With the larger and deeper imaging surveys expected in the next generation, this approach could provide sufficient precision to distinguish between different progenitor populations and shed light on the astrophysical origin of FRBs.

A Appendix

A.1 FRB Signal Compared to a Mobile Phone on the Moon

An order-of-magnitude comparison can be made using the peak flux density of an FRB and the transmitted signal of a GSM-900 mobile phone. The GSM specification gives a maximum mobile-station burst power of 2 W with a typical GSM channel spacing of 200 kHz [12]. Assuming isotropic emission at the lunar distance $d \approx 3.84 \times 10^8$ m, the received power flux is

$$F_{\text{phone}} = \frac{P}{4\pi d^2} \approx 1.08 \times 10^{-18}\,\text{W}\,\text{m}^{-2}\,. \tag{A.1}$$

If this power is distributed over one GSM channel, the phone's spectral flux density is

$$S_{\text{phone}} = \frac{F_{\text{phone}}}{B} \approx 540\,\text{Jy}\,. \tag{A.2}$$

For an FRB with $S_{\text{FRB}} \approx 0.54$ Jy, which lies within the observed range [43], the ratio is therefore $S_{\text{phone}}/S_{\text{FRB}} \approx 10^3$. This is an illustrative comparison based on the stated GSM and isotropic-emission assumptions. Other phone standards, transmit powers, and FRB flux densities yield different ratios.

A.2 LSST Y10 Forecast

This section collects the LSST Y10 inputs and results discussed in Section 5.2.

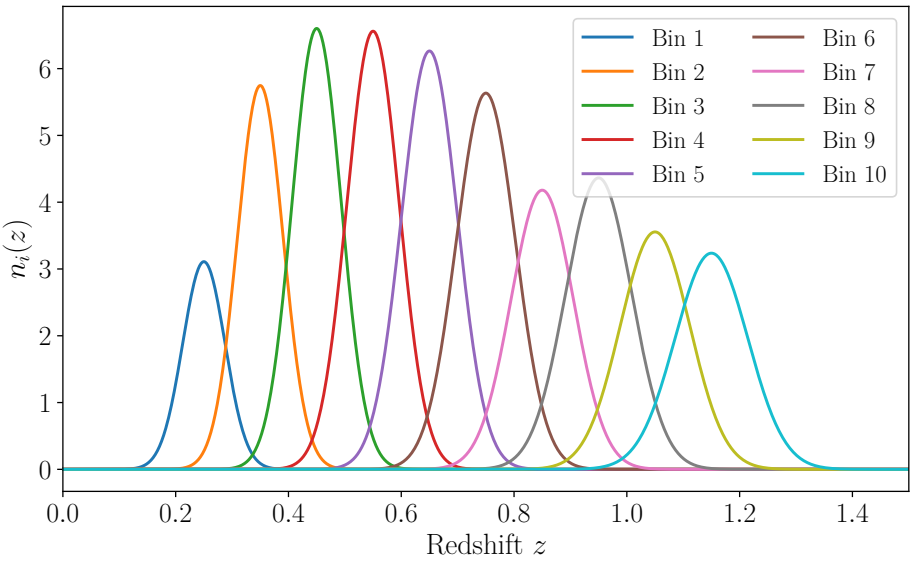
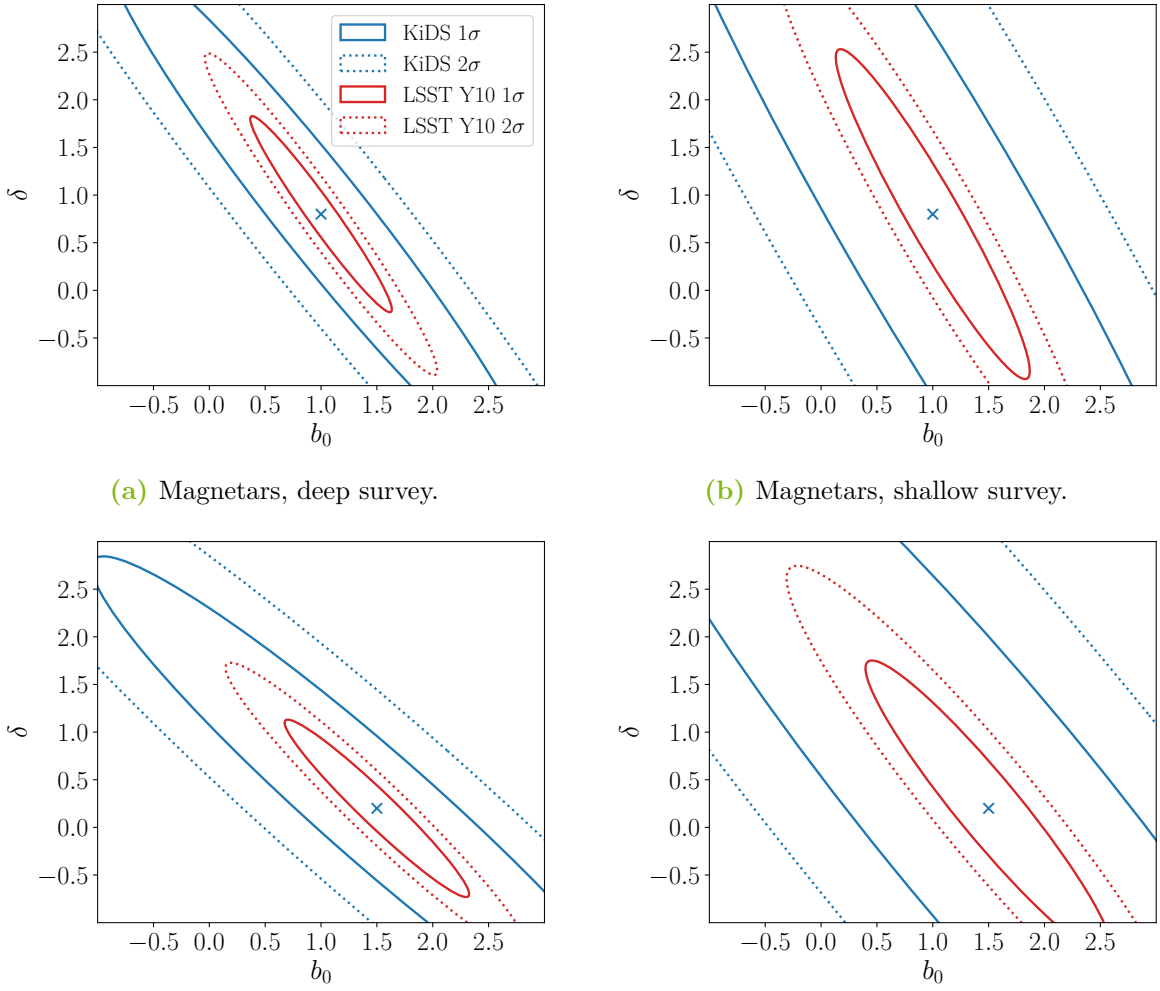


Figure A.1: Raw, tabulated tomographic redshift distributions $n_i(z)$ of the LSST Y10 lens sample as published by Zhang et al. [65].



(a) Magnetars, deep survey.

(b) Magnetars, shallow survey.

(c) Neutron stars, deep survey.

(d) Neutron stars, shallow survey.

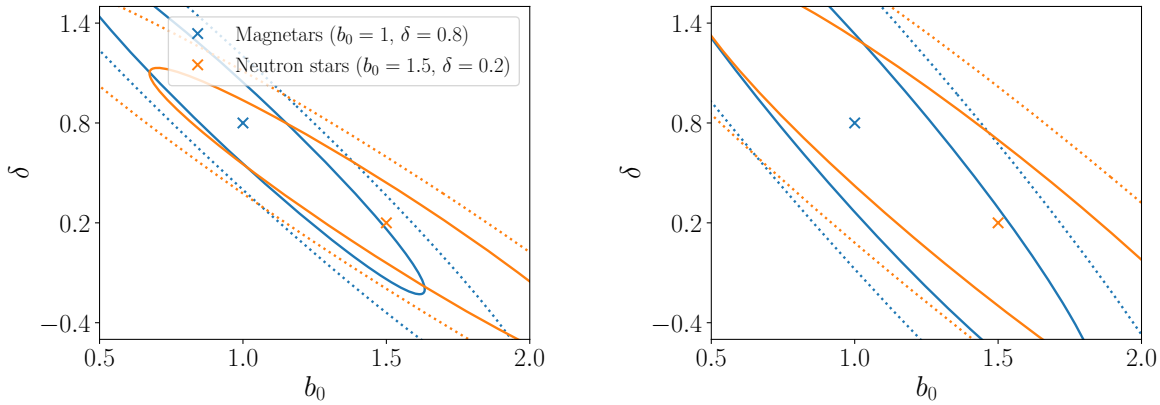
Figure A.2: Comparison of the joint FRB×galaxy forecast for the KiDS-Legacy (blue) and LSST Y10 (red) lens samples, shown for the two progenitor models and FRB survey configurations. Solid and dotted lines mark the 1σ and 2σ confidence regions, as given in the legend of panel (a). The blue cross marks the fiducial parameter values.

Table A.1: Marginalized uncertainties of the joint FRB×galaxy forecast for KiDS and LSST Y10 galaxy surveys.

		σ_{b_0}		σ_δ	
Progenitor model	Configuration	KiDS	LSST Y10	KiDS	LSST Y10
Magnetars	Deep	1.26	0.42	1.85	0.68
Magnetars	Shallow	1.90	0.57	3.73	1.14
Neutron stars	Deep	1.69	0.55	1.75	0.61
Neutron stars	Shallow	2.41	0.73	3.39	1.02

Table A.2: FoMs of the joint FRB×galaxy forecast for KiDS and LSST Y10 galaxy surveys. The last column gives the ratio of the two FoMs, computed from unrounded values.

		FoM		
Progenitor model	Configuration	KiDS	LSST Y10	Ratio
Magnetars	Deep	1.6	14.8	9
Magnetars	Shallow	0.4	4.9	12
Neutron stars	Deep	1.1	11.5	10
Neutron stars	Shallow	0.3	4.2	12



(a) Deep survey configuration.

(b) Shallow survey configuration.

Figure A.3: Joint FRB×galaxy forecast in the $b_0 - \delta$ plane for the deep (a) and shallow (b) FRB survey configuration, combined with the ten tomographic bins of the LSST Y10 lens sample. The contours of the magnetar model are shown in blue and those of the neutron-star model in orange. Solid lines mark the 1σ and dotted lines the 2σ confidence regions of the forecast. The crosses indicate the fiducial values of each model as given in the legend of panel (a). In the deep configuration, the neutron-star fiducial lies just outside the 1σ confidence contour of the magnetar model.

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